



OAKLAND COMMUNITY COLLEGE®

ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

ROB 2140 Studio 5000 LADDER LOGIC LABS

John Sefcovic
Oakland Community College

TABLE OF CONTENTS

Studio 5000 LADDER LOGIC LABS

LAB: MOVE INSTRUCTION, PART 1	1
LAB: MOVE INSTRUCTION, PART 2	5
LAB: TIMED PROCESS OUTPUT	10
LAB: TEMPERATURE CONVERSION with INDICATORS	15
LAB: BORE CYCLE	25
LAB: ANALOG INPUT	28
LAB: TORQUE	36

© 2025 John Sefcovic and Oakland Community College. This work is intended for educational purposes. The author and publisher make no express or implied warranty of any kind, nor shall they be liable in any event for any type of damage or injury, in connection with or arising out of these materials. Studio 5000 Logix Designer, Studio 5000® Logix Emulate, FactoryTalk ME, and RSLinx are registered trademarks of Rockwell Automation, Inc. of Milwaukee, Wisconsin. All such names are used only in an editorial fashion to the benefit of the trademark owner without any intention of trademark infringements. This work may be reproduced and redistributed, in whole or in part, without alteration and without prior written permission, solely by educational institutions for nonprofit administrative or educational purposes provided all copies contain this copyright statement and the Oakland Community College logo. This work is reproduced and distributed with the permission of John Sefcovic. No other use is permitted without the express prior written permission of John Sefcovic or Oakland Community College.

LAB: MOVE INSTRUCTION, PART 1

PROGRAM FILE: MOVE

OBJECTIVE: Manipulating integer data. Use programming to change the preset of a timer instruction.

PART 1 must be submitted first, then PART2.

The lab is in two parts. Create PART 1 first.

NO CREDIT WILL BE GIVEN IF PART 1 IS NOT SUBMITTED FIRST.

PART 1 - Create the Initial Program

THE MOVE INSTRUCTION MUST BE USED TO CHANGE THE TIMER PRESET.

The operator has three preset time values, 10, 15, or 20 seconds, that can be selected depending on the type of part being produced on the machine.

The operator turns a three-position selector switch, making the contact to change the preset value. The operator then presses and releases the START pushbutton, and the process output will remain on for the time period selected.

I/O CONFIGURATION:

Slot 2 - Studio Emulate 5000 Controller

Slot 3 - 1756-IB16

Slot 4 - 1756-OB16E

INPUTS:

Alias Address	Controller Tag	Descriptions
Local:3:I.Data.0	SEL_SW_1	Selector Switch, Position 1
Local:3:I.Data.1	SEL_SW_2	Selector Switch, Position 2
Local:3:I.Data.2	SEL_SW_3	Selector Switch, Position 3
Local:3:I.Data.14	START	START Momentary N. O. Pushbutton



OUTPUT:

Alias Address	Controller Tag	Descriptions
Local:4:O.Data.0	OUTPUT	Process Output

INTERNALS:

Param/Local Tag	Data Type	Descriptions
PRE_SET_1	Integer	Preset No.1
PRE_SET_2	Integer	Preset No.2
PRE_SET_3	Integer	Preset No.3
ONE_SHOT	Single Integer Array	One Shot Control Array

Entry of Single Integer Array, **Data Type** SINT[1].

The array consists of a Single Integer, eight bits, of one word.
If a Boolean array is used, by default, must be in increments of 32 bits.

Name	Usage	Alias	Base Tag	Data Type
▶ DWELL	Local			TIMER
▶ ONS_SHOT	Local			SINT[1]
▶ PRE_SET_1	Local			INT
▶ PRE_SET_2	Local			INT
▶ PRE_SET_3	Local			INT

Select the **Monitor Tab** and enter the Pre_Set Tag **Values**.

Monitor Tags

Name	Usage	Value	Force Mask	Style	Data Type
▶ DWELL	Local	{...}		{...}	TIMER
▶ ONS_SHOT	Local	{...}		{...}	SINT[1]
▶ PRE_SET_1	Local	10000		Decimal	INT
▶ PRE_SET_2	Local	15000		Decimal	INT
▶ PRE_SET_3	Local	20000		Decimal	INT

TIMER:

Param/Local Tag	Data Type	Descriptions
DWELL	Timer	Process Timer
DWELL.PRE*	Double Integer	Process Timer Preset
DWELL.ACC*	Double Integer	Process Timer Accumulated
DWELL.EN*	Boolean	Process Timer Enabled
DWELL.DN*	Boolean	Process Timer Complete

* Created automatically in DWELL timer structure.



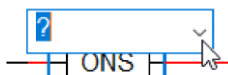
Entering One-Shot

The One-Shot, [ONS], instruction is in the **Bit** tab on the Instruction Palette.



After dragging the instruction onto the rung the Control Bit is assigned.

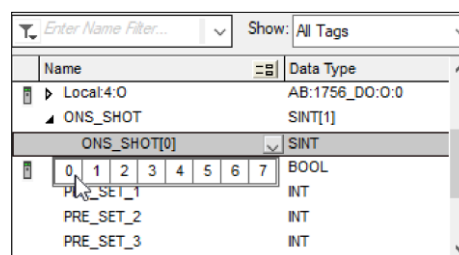
Select the down arrow in the Tag field.



Expand the Tag ONE_SHOT.

Select the down arrow in the ONE_SHOT[0] Word.

Select the Bit number for the table.



Each One-Shot instruction in the program [must have a unique Bit Number assigned](#).

DOWNLOAD

Download the PLC program and place the PLC into RUN

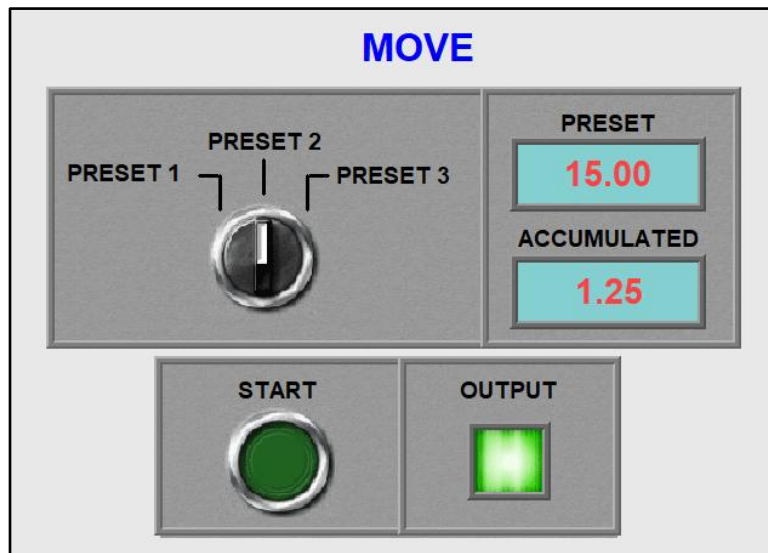
*An HMI has been created to allow you to test the logic, **MOVE_TEST_HMI**.*

The HMI is a runtime file.

See desktop PDF document [FactoryTalk View Studio Machine Edition Procedures](#) on page 15 [Appendix B: STATION RUNTIME APPLICATION](#).

continue

TESTING



1. Left click on the Selector Switch and select **PRESET 1**.
2. Press the **START** button.

The output indicator for the **OUTPUT** status should be ON for 10 seconds.

1. Left click on the Selector Switch and select **PRESET 2**.
2. Press the **START** button.

The output indicator for the **OUTPUT** status should be ON for 15 seconds.

1. Left click on the Selector Switch and select **PRESET 3**.
2. Press the **START** button.

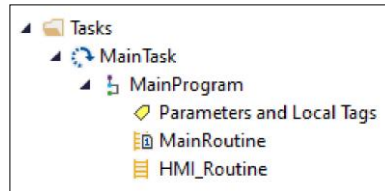
The output indicator for the **OUTPUT** status should be ON for 20 seconds.

LAB: MOVE INSTRUCTION, PART 2

PROGRAM FILE: MOVE *created for PART 1*

OBJECTIVE: Manipulating integer data. Use programming to change the preset of a timer instruction. Adding logic considerations for HMI.

Routine Structure



PLC and HMI DEVELOPMENT

Create the PLC program described below.

Then, create the HMI program for the template. See [ROB 2140 HMI Labs](#).

Download the PLC program and place the PLC into RUN and execute the HMI in in Design Testing.

Test the PLC program with the HMI. If there are problems determine if the problem is with the PLC's logic or the HMI settings.

PART 2 - Add Subroutine and Modifications for HMI

The values are selected via the Operator's Human Machine Interface (HMI) panels. A START button on the HMI turns on the process output for the selected time period. The current preset and accumulated values are displayed. A second panel allows the three preset values to be entered.

As a backup a hardwired operators panel can be used when not disabled by the HMI.

Create a new routine HMI operator's panel logic.

continue

Name of Routine: HMI_Routine.

Add to the program's **HMI_Routine** the following Param/Local Tags.

OPERATOR'S PANEL:

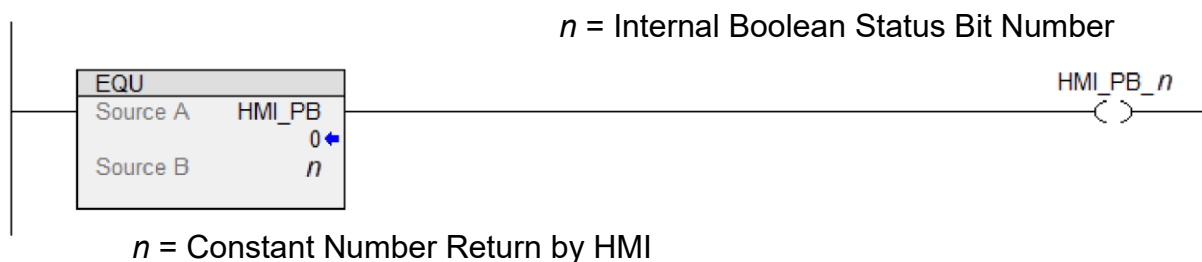
Param/Local Tag	Data Type	Descriptions
HMI_PB	Single Integer	Operator Panel Pushbuttons
HMI_PB_1	Boolean	Operator Panel Pushbutton 1
HMI_PB_2	Boolean	Operator Panel Pushbutton 2
HMI_PB_3	Boolean	Operator Panel Pushbutton 3
HMI_ENABLE_DISABLE	Boolean	Allows Hardwired Panel
HMI_PRE	Real	Operator Panel Timer Preset
HMI_ACC	Real	Operator Panel Timer Accumulated

In this routine create the rungs:

The HMI_PB tag returns the value of the Operator Pushbutton pressed on the HMI.

The EQU instructions sets correspondingly, the status bit HMI_PB_1 when OP_PB value is 1, set status bit HMI_PB_2 when HMI_PB value is 2, and set status bit HMI_PB_3 when HMI_PB value is 3.

Three rungs are needed.

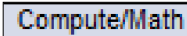


Use an Arithmetic Instruction on a new rung to change the preset DWELL timer's .PRE value to represent the preset in actual seconds for the operator panel timer preset. The preset .PRE is in milliseconds (default time base). Divide the timer's .PRE by 1000 and place the result into the Tag for displaying on the HMI.



Use an Arithmetic Instruction on a new rung to change the preset DWELL timer's .ACC value to represent the preset in actual seconds for the operator panel timer preset. The accumulated .ACC is in milliseconds (default time base). Divide the timer's .ACC by 1000 and place the result into the Tag for displaying on the HMI.

The Divide (**DIV**)  instruction is in the Instruction Palette **Compute/Math** Tab.



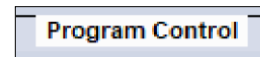
MainRoutine

Modification 1

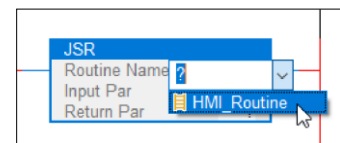
Add a Rung at the top of the program, new Rung 0.

 The HMI_Program routine must be created first.

The Jump to Subroutine **JSR** is in the **Program Control** tab.

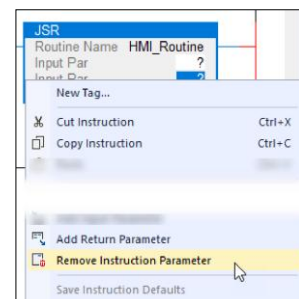


The *new* first rung is the unconditional Jump to Subroutine instruction.



Double left click on the Routine Name and select the routine from the drop-down list.

Right click on the instruction for the context menu and remove all the instruction's Input Par and Output Par parameters.



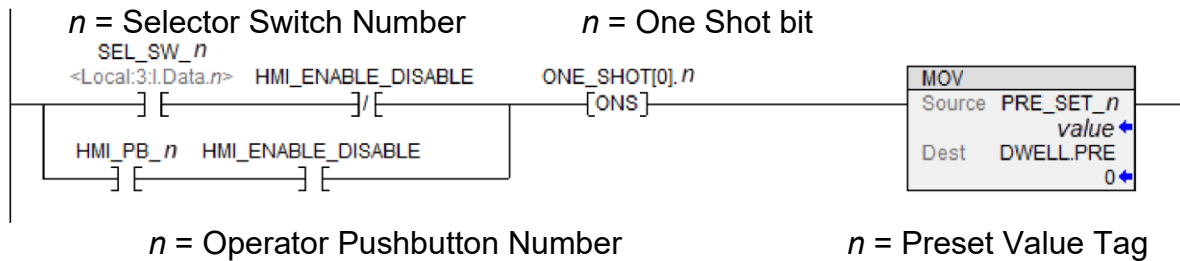
Modification 2

Modify the switch rungs to move the selected preset value with either the Selector Switch or the Operator's HMI Panel.

All three of the MOVE rungs will require these modifications.

The selector switch will only be operational if the HMI_ENABLE_DISABLE on the operators panel is low.

Note - Information Only: This condition for HMI_ENABLE_DISABLE SWITCH low would exist if the HMI panel was not operational.



DOWNLOAD

Test the PLC program with the HMI. If there are problems determine if the problem is with the PLC's logic or the HMI settings.

Test using the HMI you created for the HUMAN MACHINE INTERFACE (HMI) LABS the MOVE INSTRUCTION lab.

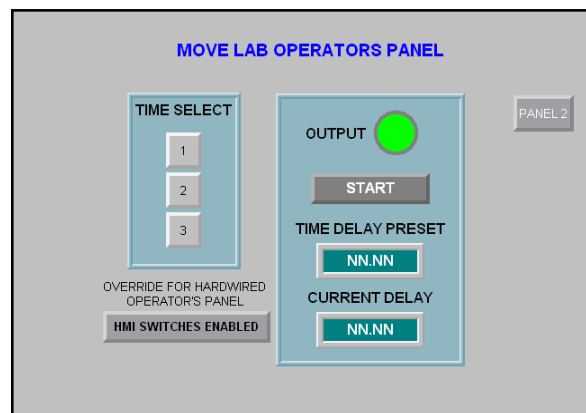
Download the PLC program and place the PLC into RUN and execute the HMI in in Design Testing.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 10 TESTING APPLICATION IN DESIGN.

TESTING

1. Press the button for **TIME SELECT 1**.
2. Press the **START** button.

The output indicator for the **OUTPUT** status should be ON for 10 seconds.



continue

1. Press the button for **TIME SELECT 2**.

2. Press the **START** button.

The output indicator for the OUTPUT status should be ON for 15 seconds.

1. Press the button for **TIME SELECT 3**.

2. Press the **START** button.

The output indicator for the OUTPUT status should be ON for 20 seconds.

LAB: TIMED PROCESS OUTPUT

PROGRAM FILE: PROCESS_OUTPUT

OBJECTIVE: Arithmetic instructions and Floating Point data type.

Create a program for a process that uses multiple layers of a sheet material. The type and number of sheets loaded depends on the order. The operator enters one of four types of sheet materials that will be used and the number of sheet layers. Only one type of material is used at a time, and up to six sheets may be used.

The operator presses an HMI panel pushbutton that corresponds to the material type, then presses an HMI panel pushbutton for the number of sheets. When the operator presses and releases the START pushbutton, a process output will turn on for the computing determined time value based on the type and number of sheets. The following parameters are used to control the process timer.

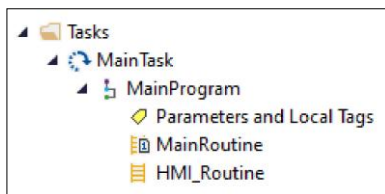
Material A - multiplier of 0.73 times the number of sheets

Material B - multiplier of 1.05 times the number of sheets

Material C - multiplier of 1.23 times the number of sheets

Material D - multiplier of 1.86 times the number of sheets

Routine Structure



I/O CONFIGURATION:

Slot 2 - Studio Emulate 5000 Controller

Slot 3 - 1756-IB16

Slot 4 - 1756-OB16E

INPUTS:

Alias Address	Controller Tag	Descriptions
Local:3:I.Data.0	SW_START	START Momentary N. O. Pushbutton

OUTPUT:

Alias Address	Controller Tag	Descriptions
Local:4:O.Data.0	PROCESS_OUTPUT	Process Output



TIMER:

Param/Local Tag	Data Type	Descriptions
PROCESS_TIME	Timer	Process Timer

INTERNALS:

These Tags are used in the PLC logic program only.

Param/Local Tag	Data Type	Descriptions
A_MATERIAL_MULTIPLIER	Real	Material A Multiplier
B_MATERIAL_MULTIPLIER	Real	Material B Multiplier
C_MATERIAL_MULTIPLIER	Real	Material C Multiplier
D_MATERIAL_MULTIPLIER	Real	Material D Multiplier
NUM_SHEETS	Integer	Value for Sheet Type
TYPE_SHEET	Integer	Value Number of Sheets
MATERIAL_MULTI_VALUE	Real	Material Multiplier Value
PROCESS_VALUE	Real	Process Value

Alias for HMI:

These Tags are for use in the HM only.

Param/Local Tag	Alias For
HMI_A_MATERIAL_MULTIPLIER	A_MATERIAL_MULTIPLIER
HMI_B_MATERIAL_MULTIPLIER	B_MATERIAL_MULTIPLIER
HMI_C_MATERIAL_MULTIPLIER	C_MATERIAL_MULTIPLIER
HMI_D_MATERIAL_MULTIPLIER	D_MATERIAL_MULTIPLIER
HMI_NUM_SHEETS	NUM_SHEETS
HMI_TYPE_SHEET	TYPE_SHEET
HMI_PRE_PROCESS_TIME	PROCESS_TIME.PRE
HMI_ACC_PROCESS_TIME	PROCESS_TIME.ACC

The **Alias For** identifies these tags for use with the HMI.

continue



The example below only illustrates the use of an Alias For tag name for the HMI_ACC_PROCESS_TIME and HMI_PRE_PROCESS_TIME.

*Notice the **Alias For** is the **PROCESS_TIMER**'s Preset and Accumulated values. A Tag from an Alias provides for a more descriptive renaming of a duplicate tag.*

Name	Usage	Alias For	Base Tag	Data Type
▶ HMI_NUM_SHEET	Local			INT
▶ HMI_TYPE_SHEET	Local			INT
▶ PROCESS_TIMER	Local			TIMER
▶ PROCESS_TIMER.ACC				DINT
PROCESS_TIMER.DN				BOOL
PROCESS_TIMER.EN				BOOL
▶ PROCESS_TIMER.PRE				DINT
PROCESS_TIMER.TT				BOOL
PROCESS_VALUE	Local			REAL
▶ HMI_ACC_PROCESS_TIME	Local	PROCESS_TIMER.ACC	PROCESS_TIMER.ACC	DINT
▶ HMI_PRE_PROCESS_TIME	Local	PROCESS_TIMER.PRE	PROCESS_TIMER.PRE	DINT

The **Alias For** concept applies to the other Tags in the assignment to create duplicate data descriptive Tag Names for the HMI.

LOGIC FOR MainRoutine

An unconditional call of the subroutine for HMI panel.

The pressing of the START pushbutton calculates by using a Multiply instruction for the NUM_SHEETS by the MATERIAL_MULTI_VALUE for the product result of the PROCESS_VALUE.

A second Multiply instruction, controlled by the START, converts the PROCESS_VALUE to milliseconds for Preset Time of the Timer.

The START then turns on the Output and begins the timer elapsing for the Process Value time. To then turn off the Output at the completion of the timing cycle

LOGIC FOR OPERATOR'S HMI PANEL

Name of Routine: HMI_Routine

Four similar Rungs use a Compare instructions determines the four values of the TYPE_SHEET to be equal to HMI Interlock button value and moves the selected A_MATERIAL_MULTIPLIER, B_MATERIAL_MULTIPLIER, C_MATERIAL_MULTIPLIER, or D_MATERIAL_MULTIPLIER into the MATERIAL_MULTI_VALUE.

continue

DOWNLOAD

Test the PLC program with the HMI. If there are problems determine if the problem is with the PLC's logic or the HMI settings.

Test using the HMI you created for the HUMAN MACHINE INTERFACE (HMI) LABS the TIMED PROCESS OUTPUT lab.

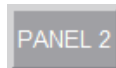
Download the PLC program and place the PLC into RUN and execute the HMI in in Design Testing.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 10 TESTING APPLICATION IN DESIGN.

TESTING

ENTERING MATERIAL MULTIPLER

1. Press the button to go to **PANEL 2**.



2. Enter the values for the **A, B, C, and D MATERIALS**.

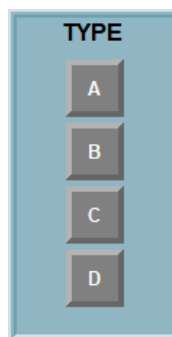
3. Press the button to return to the **MAIN** screen.



MULTIPLIER VALUE SET-UP	
MATERIAL A	0.73
MATERIAL B	1.05
MATERIAL C	1.23
MATERIAL D	1.86

SELECTING SHEET TYPE AND NUMBER

1. Press the Button for **TYPE B**.



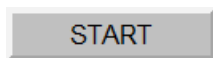
continue



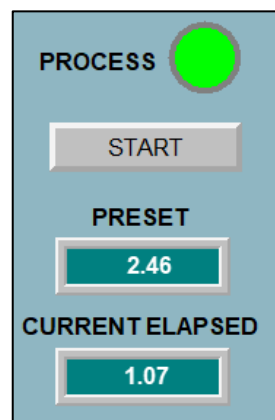
2. Press the Button for **NUMBER 2**.



3. Press the **START** button.



The **OUTPUT** should be ON while the timer is **ELASPING**.



4. Select the other **TYPE** and **NUMBER** buttons and then **START** to verify the different values calculated values for the Timer's **PRESET**.

LAB: TEMPERATURE CONVERSION with INDICATORS

PROGRAM FILE: CONVERT

OBJECTIVE: Comparison instructions and the buffering of data. Comparison instruction to determine a data range

Create a program that will convert a Celsius temperature to a Fahrenheit temperature value. The maximum Celsius value will be 100 degrees.

I/O CONFIGURATION:

Slot 2 - Studio Emulate 5000 Controller

Slot 3 - 1756-IB16

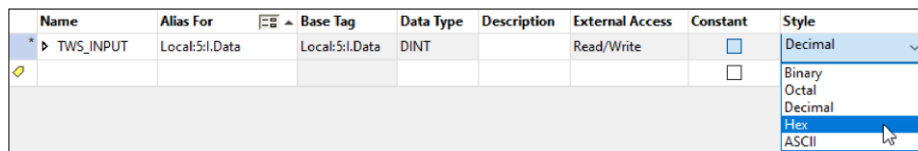
Slot 4 - 1756-OB16E

Slot 5 - 1756-IB16

NOTE: TWS_INPUT is a Double Integer Data Type and addresses the entire data word. Notice no **.bit** is used in the Alias Address. The data is shown in a HEX Style.

INPUT:

Alias Address	Controller Tag	Descriptions
Local:5:I.Data	TWS_INPUT	Thumbwheel Switch, Celsius Value
Local:3:I.Data.14	CONVERT_PB	Convert Temperature N. O. pushbutton
Local:3:I.Data.15	RESET_PB	Data Error Reset, N. O. pushbutton



TWS_INPUT will be in Hexadecimal. In the **Style** column in the TWS_INPUT row.

Left click and select **Hex** from the drop-down list. Hex style number display format is similar to the BCD format.

OUTPUT:

Alias Address	Controller Tag	Descriptions
Local:4:O.Data.0	ERROR_DUP_VAL	Error Duplicate Value Indicator
Local:4:O.Data.1	ERROR_OVER_VAL	Data Error, Over Value

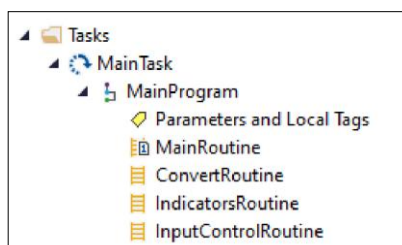


Local:4:O.Data.2	LL_INDC	Low Low Indicator
Local:4:O.Data.3	L_INDC	Low Indicator
Local:4:O.Data.4	M_INDC	Mid Indicator
Local:4:O.Data.5	H_INDC	High Indicator
Local:4:O.Data.6	HH_INDC	High High Indicator

INTERNALS:

Param/Local Tag	Data Type	Description
ONE_SHOT_CONTROL	Boolean	One Shot
INTERNAL_CONVERT	Boolean	Internal Control, Enable Convert
C_TEMP	DINT	Celsius Temperature
C_TEMP_STORAGE	DINT	Storage of Celsius Temperature
C_TEMP_CONVERT	DINT	Celsius Temperature Parameter
F_TEMP	DINT	Fahrenheit Value
F_TEMP_CONVERT	DINT	Fahrenheit Temperature Parameter
MUL_RESULT	DINT	Multiply Result
DIV_RESULT	DINT	Divide Result
HMI_CONVERT_PB	Boolean	HMI Convert Button
HMI_RESET_PB	Boolean	HMI Reset Button
HMI_ENABLE_DISABLE	Boolean	HMI Enable/Disable Button

ROUTINE STRUCTURE



continue



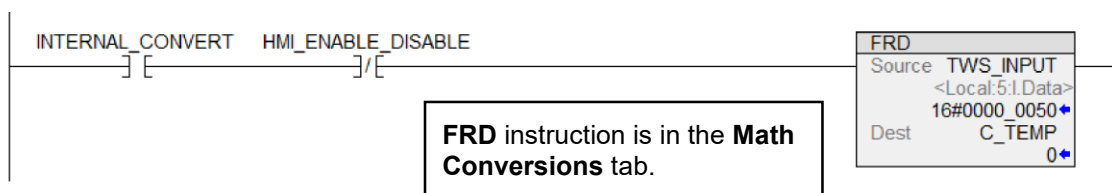
Name of Routine: InputControlRoutine

NOTE: For the one-shot instruction, use the **ONS**. The ONS is in the **Bit** tab.

NOTE: When using the JSR, SBR, and RET instructions, delete unused parameters by right clicking on the parameter and selecting Remove Instruction Parameter.

Note: The CONVERT TEMPERATURE and RESET pushbuttons (switches) will only be operational if HMI ENABLE DISABLE is reset (logic 0).

If Enabled, the temperature to be converted will be entered via a thumbwheel switch as a BCD value.



If the thumbwheel switch is Disabled the value the Celsius temperature (C_TEMP) originates from the HMI.

When the operator presses the CONVERT TEMPERATURE pushbutton, the program will perform the following error checks.

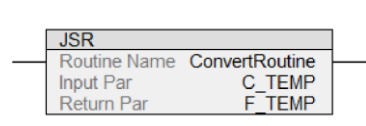
If the value entered has not changed, a duplicate error indicator will be lit. A check for the input temperature is 100 degrees or less will be performed.

If it is above 100 degrees, the Over Value indicator will be lit.

Any error will prevent a conversion input value.

The operator must then press the RESET pushbutton and enter the correct value.

If there are no errors, then the Input Parameter of the Celsius temperature is passed to the called **Routine: ConvertRoutine**. The Fahrenheit temperature is placed in the Return Parameter.



continue

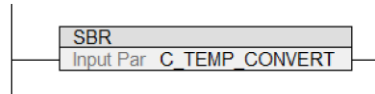
Name of Routine: ConvertRoutine

Use the instructions in the **Compute** tab to convert a Celsius temperature to a Fahrenheit temperature value.

Equation: $F = (C(9)/5) + 32$ You **Cannot** use the CPT instruction.

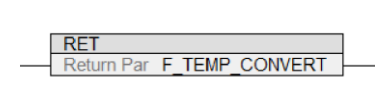
Note: the **SBR** and **RET** instructions are in the **Program Control** tab.

The passed tag C_TEMP is placed into the subroutine tag C_TEMP_CONVERT.



The Celsius temperature tag C_TEMP_CONVERT is converted using the Compute/Math instructions and the resulting Fahrenheit temperature is placed in the tag F_TEMP_CONVERT.

The F_TEMP_CONVERT is returned to the F_TEMP tag.



The **RET** can be the only instruction on the rung.

Name of Routine: IndicatorsRoutine

The **Limit Test** instruction for the range indicators **must** be used.

If a conversion is valid, indicators will show the converted temperature is within a LOW LOW, LOW, MID, HIGH, or HIGH HIGH range. Use the LIM instructions.

Ranges:	Low Low	32 to 49
	Low	50 to 99
	Mid	100 to 149
	High	150 to 199
	High High	200 to 212

Name of Routine: MainRoutine

The Routine **InputControlRoutine** for the conversion is called unconditionally.
The Routine **IndicatorsRoutine** for the indicators is called unconditionally.

DOWNLOAD

Download the program to the PLC.

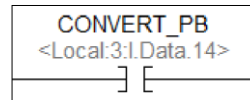
See Procedures below to **Test Without the HMI**.

TESTING WITHOUT THE HMI

PROCEDURES TO TEST THE PROGRAM WITHOUT THE HMI.

PROCEDURE - TO CHANGE PUSHBUTTON STATUS IN EMULATION ONLY

1. Select the **Convert Pushbutton**.
2. Press and hold **Ctrl** and then press **T**.
3. Press and hold **Ctrl** and then press **T** to toggle back to original status.



*When needed use the **Ctrl - T** procedure to test the **Reset pushbutton**.*

*When needed use the **Ctrl - T** procedure to test the **Enable/Disable thumbwheel switch**.*

PROCEDURE - TO ENTER A CELSIUS VALUE IN STUDIO 5000:

*Use this procedure to test the values for the **Temperature Ranges**, and **Duplicate Value** and **Over Value errors**.*

1. Open the **Parameters and Local Tags** table.
2. Select the **Monitor** tab.
3. Select the **Value** column field for the **C_TEMP** tag.

Name	Usage	Value	Force Mask	Style
▸ C_TEMP	Local	60		Decimal

4. Enter a Celsius value and press **Enter**. *Type in 0 for initial value.*
-

continue

PROCEDURE - TO ENTER A THUMBWHEEL SWITCH CELSIUS VALUE

1. Open the **Controller Tags** table.
2. Select **Monitor Tags** tab.
3. Select the **Value** column field for the **TWS_INPUT** tag.

Name	Value	Force Mask	Style
▶ TWS_INPUT	16#0000_0050		Hex

4. Enter a Celsius value (example 50) and press **Enter**.

TESTING CELSIUS TO FAHRENHEIT CONVERSION WITHOUT HMI

Select the **InputControlRoutine**.

See above procedure to **CHANGE PUSHBUTTON STATUS**.

1. Toggle **HMI_ENABLE_DISABLE** to be True (Logic 1).
2. Set a 0 value for the **C_TEMP** tag.
3. Toggle (ON-OFF) the **CONVERT_PB** pushbutton
4. Check for the correct converted temperature, **should be 32° F**.
5. Check for the correct **Range Indicator**.

Repeat the steps 2, 3, 4, and 5 with the other Celsius values into C_TEMP to test conversion and the Range Indicators.

Celsius	Fahrenheit	Range
0°	32°	Low Low
30°	86°	Low
60°	140°	Mid
80°	176°	High
100°	212°	High High

TESTING DUPLICATE ERROR WITHOUT HMI

1. Enter a Celsius value of 40 into C_TEMP.
2. Toggle (ON-OFF) the **CONVERT_PB** pushbutton.
3. Again, toggle (ON-OFF) the **CONVERT_PB** pushbutton.
ERROR_DUP_VAL indicator should be ON.
4. Toggle (ON-OFF) the **RESET_PB** pushbutton.
ERROR_DUP_VAL indicator should be OFF.



TESTING OVER TEMPERATURE VALUE ERROR WITHOUT HMI

1. Enter a Celsius value of 101 into C_TEMP.
2. Toggle (ON-OFF) the **CONVERT_PB** pushbutton.
3. Again, toggle (ON-OFF) the **CONVERT_PB** pushbutton.
ERROR_OVER_VAL indicator should be ON.
4. Toggle (ON-OFF) the **RESET_PB** pushbutton.
ERROR_OVER_VAL indicator should be OFF.

TO ENTER BCD THUMBWHEEL VALUE FOR TESTING WITHOUT HMI

YOU WILL BE REQUIRED TO DEMONSTRATE THE THUMBWHEEL SWITCHES WHEN THE HMI IS GRADED.

*Procedure for testing TWS_INPUT value when HMI is **DISABLED**.*

1. Open the **Controller Tags** table.
2. Select **Monitor Tags** tab.
3. Select the **Value** column field for the **TWS_INPUT** tag.

Name	Value	Force Mask	Style
▸ TWS_INPUT	16#0000_0050		Hex

4. Enter a Celsius value 50 and press **Enter**.
5. Toggle **HMI_ENABLE_DISABLE** to be False (Logic 0).
6. Toggle (ON-OFF) the **CONVERT_PB** pushbutton
7. Check for the correct converted temperature, *should be 122° F.*

TESTING WITH THE HMI

Test the PLC program with the HMI. If there are problems determine if the problem is with the PLC's logic or the HMI settings.

To Test use the HMI you created for the HUMAN MACHINE INTERFACE (HMI) LABS the TEMPERATURE CONVERSION with INDICATORS lab.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 15 Appendix B: STATION RUNTIME APPLICATION.

continue



ENABLE HMI

1. From the MAIN screen select the **MAINTENANCE** button.



2. In the MAINTANCE screen select **TWS DISABLE**.



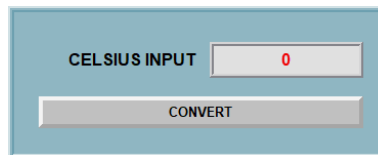
3. Select the go to **INPUT** button.



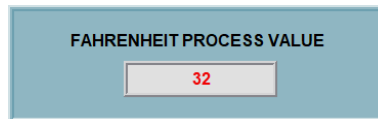
TESTING CELSIUS TO FAHRENHEIT CONVERSION

1. Set a 0 value for the **CELSIUS INPUT**.

2. Press the **CONVERT** pushbutton.



3. Check for the correct converted temperature, [should be 32° F](#).



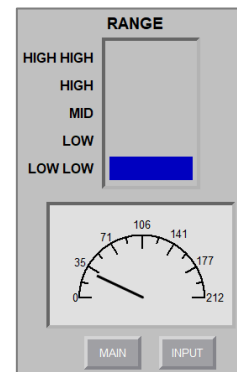
4. Press the **RANGE** button.



4. Check for the correct **Range Indicator and Gauge** value.

[Repeat the steps 1 through 7 with the other Celsius values into C_TEMP to test conversion and the Range Indicators.](#)

Celsius	Fahrenheit	Range
0°	32°	Low Low
30°	86°	Low
60°	140°	Mid
80°	176°	High
100°	212°	High High

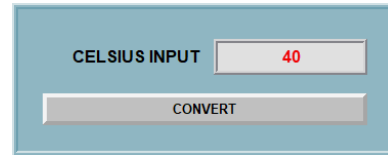


continue

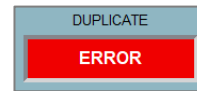


TESTING DUPLICATE ERROR

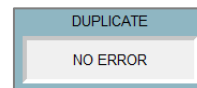
1. Enter a Celsius value of 40 into **CELSIUS INPUT**.
2. Press the **CONVERT** pushbutton.



3. Again, press the **CONVERT** pushbutton.
DUPLICATE ERROR indicator should be ON.

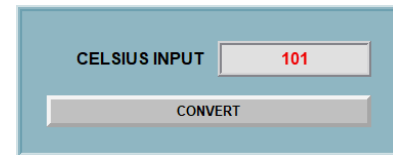


4. Press the **RESET** pushbutton.
DUPLICATE ERROR indicator should be OFF.



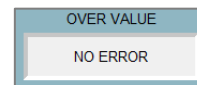
TESTING OVER TEMPRATURE VALUE ERROR WITHOUT HMI

1. Enter a Celsius value of 101 into **CELSIUS INPUT**.
2. Press the **CONVERT** pushbutton.



OVER VALUE ERROR indicator should be ON.

3. Toggle (ON-OFF) the **RESET_PB** pushbutton.
OVER VALUE ERROR indicator should be OFF.



YOU WILL BE REQUIRED TO DEMONSTRATE THE THUMBWHEEL SWITCHES WHEN THE HMI IS GRADED.

ENABLE HMI

1. Select the go to **MAIN** button.
2. Select the **MAINTENANCE** button.
3. Change to **TWS ENABLED**.



4. Select the go to **INPUT** button.



continue



TO ENTER BCD THUMBWHEEL VALUE ONLINE IN STUDIO 5000

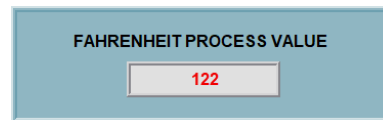
1. Open the **Controller Tags** table.
2. Select **Monitor Tags** tab.
3. Select the **Value** column field for the **TWS_INPUT** tag.

Name	Value	Force Mask	Style
▶ TWS_INPUT	16#0000_0050		Hex

4. Enter a Celsius value 50 and press **Enter**.
5. Press the HMI **CONVERT** pushbutton.



6. Check for the correct converted temperature, [should be 122° F](#).



LAB: BORE CYCLE**PROCESS MODEL: BORE**

OBJECTIVE: Use of internal Boolean data types as control relays to sequence the operations of a machine.

NOTE: The Outputs and START input are already in the program

The operator loads a part into the boring machine; a normally open photo-electric switch will monitor for the part in the correct position. The operator will then close the Safety Gate. A normally open switch on the Safety Gate will be closed and the safety controller will send a positive signal to the PLC the gate is closed. If the Safety Gate opens at any time, the Spindle and the Up or Down Solenoids will turn off.

Operator presses and holds the Start Pushbuttons:

When the SAFETY GATE is closed, the operator turns the ENGAGE CLAMPS normally open maintained switch, the two clamp solenoids are energized. A normally open limit switch on each clamp monitor for a complete engagement.

When the clamps are engaged and the safety gate is closed, the operator then presses and holds the anti-tie down start pushbuttons, the safety controller will send positive signal to the PLC to START the cycle.

The positive START signal energizes the Spindle Motor and Down Solenoid to lower the spindle carriage.

The pushbuttons are pressed for the entire boring operation.

When the spindle travel hits the normally open DOWN LIMIT SWITCH, the SPINDLE MOTOR and DOWN SOLENOID will be de-energized, and the UP SOLENOID will be energized. The UP SOLENOID will be de-energized when the spindle carriage hits the normally open UP LIMIT SWITCH.

Operator releases one of the Start Pushbuttons:

If the operator releases either of the pushbuttons: the Up or Down solenoid that is energized will be turned off, and if the spindle is running, it will be turned off. The hydraulic solenoids for up and down are ported so that if the solenoid turns off, the spindle carriage will remain stationary.

Not used in this program are switches that would allow for manually moving the spindle carriage up or down.

The program must resume the last action of the SPINDLE MOTOR and the UP or DOWN SOLENOIDS of the spindle carriage when the start pushbuttons are pressed.

INPUTS:

Alias Address	Controller Tag	Descriptions
Local:3:I.Data.0	PART_PE	Part in Position Photo-electric
Local:3:I.Data.1	ENGAGE_CLAMP_SW	Engage Clamps Two-position Switch
Local:3:I.Data.2	LS_DOWN	Down Limit Switch, N. O.
Local:3:I.Data.3	LS_UP	Up Limit Switch, N. O.
Local:3:I.Data.10	SAFETY_GATE	Safety Gate, N. O. Switch
Local:3:I.Data.11	LS_CLAMP1_ENGAGED	Clamp No. 1 Engaged Limit Switch
Local:3:I.Data.12	LS_CLAMP2_ENGAGED	Clamp No. 2 Engaged Limit Switch
Local:3:I.Data.14	START_SIGNAL	Start No. 1 Pushbutton

OUTPUTS:

Alias Address	Controller Tag	Descriptions
Local:4:O.Data.0	SPINDLE_ON	Spindle Motor On
Local:4:O.Data.1	SOL_DOWN	Down Solenoid
Local:4:O.Data.2	SOL_UP	Up Solenoid
Local:4:O.Data.10	SOL_CLAMP1	Clamp No. 1 Solenoid
Local:4:O.Data.11	SOL_CLAMP2	Clamp No. 2 Solenoid

INTERNALS:

Param/Local Tag	Data Type	Description
CYCLE_START_CR	BOOL	Cycle Started Control Relay
CYCLE_IN_PROGRESS_CR	BOOL	Cycle is in Progress Control Relay
DOWN_CR	BOOL	Down Control Relay
UP_CR	BOOL	Up Control Relay
CYCLE_COMPLETE_CR	BOOL	Cycle Completed Control Relay

continue

DOWNLOAD

Download the PLC program and place the PLC into RUN

*An HMI has been created to allow you to test the logic, **BORE_HMI**.*

The HMI is a runtime file.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 15 Appendix B: STATION RUNTIME APPLICATION.

There is a Desktop shortcut that has an animation describing the HMI screen and sequence of operations.



LAB: ANALOG INPUT

PROCESS MODEL: ANALOG_INPUT

OBJECTIVE: Analog input for process control.

The analog input is used to measure the height of thin stamped metal part as they are fed to an automatic assembly machine. Duplicate parts are assembled as a stacked “sandwich” before being joined. The stamped parts are picked by vacuum cups off of a stack and placed onto a conveyor. The single analog sensor measures the height of the part to make sure that only one part was placed on the conveyor.

If two parts are stuck together, the height measurement will detect the two parts. The process is stopped, and a Feed Error indicator will turn on. This indicator stays on until the Reset pushbutton is pressed.

NOTE: The Feed Error must be latched on to retain status, after problem corrected and before reset, during power failure, or change of modes.

The operator will correct the problem, and then press Reset to turn off the Error indicator. The Start pushbutton is pressed to restart the system.

Two proximity switches are used to detect the part type. The analog sensor will then measure the height of the part between 0 and 100. This is not an actual calibrated measurement, but a relative measurement based on the percentage of movement of the linear stroke of the sensor.

Part A has a nominal percentage of 30, a value above 45% will indicate two parts.

Part B has a nominal percentage of 40, a value above 55% will indicate two parts.

Use an Add On Instruction to scale the Height Sensor Analog Input. The Add On Instruction will use the language Type **Structured Text**.

I/O CONFIGURATION:

Slot 2 - Studio Emulate 5000 Controller
Slot 3 - 1756-IB16
Slot 4 - 1756-OB16E
Slot 5 - 1756-IF8

NOTE: Tags which begin with **emTagName** are for the emulation model and are not part of the program.

continue



INPUTS:

Alias Address	Controller Tag	Descriptions
Local:3:I.Data.0	PART_A	Part A N. O. Proximity Switch
Local:3:I.Data.1	PART_B	Part B N. O. Proximity Switch
Local:3:I.Data.2	RESET	RESET N. O. Pushbutton
Local:3:I.Data.14	START	START N. O. Pushbutton
Local:3:I.Data.15	STOP	STOP N. C. Pushbutton

ANALOG INPUT:

Analog Data is a REAL Data Type by default.

Alias Address	Controller Tag	Descriptions
Local:5:I.Ch0Data	HEIGHT_SENSOR	Height Sensor for Add On

OUTPUTS:

Alias Address	Controller Tag	Descriptions
Local:4:O.Data.0	CONVEYOR	Conveyor Motor
Local:4:O.Data.1	ERROR	Part Feed Error Indicator

INTERNALS:

Controller Tag	Data Type	Descriptions
PART_A_ERROR	Boolean	Part A double part
PART_B_ERROR	Boolean	Part B double part
PART_A_UPPER	Real	Part A Upper Limit
PART_B_UPPER	Real	Part B Upper Limit
HEIGHT	DINT	Scaled EU Value from Add On

ADD ON INSTRUCTION:

NOTE: Add On Instruction must be created before adding this Tag.

Controller Tag	Data Type	Descriptions
HEIGHT_SCALE	Analog_In	Add On Instruction Tag for Analog Scaling



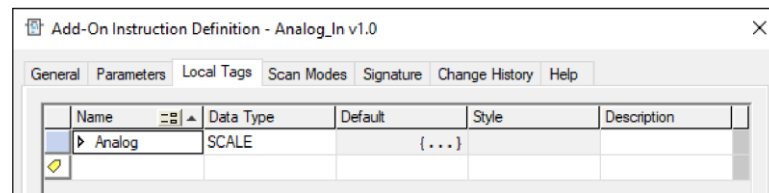
Add on Instruction:

Add On Name: Analog_In

Parameters:

Name	Usage	Data Type	Req/Vis	Descriptions
Raw_Max	Input	REAL	✓	Maximum Bit Value
Raw_Min	Input	REAL	✓	Minimum Bit Value
EU_Max	Input	REAL	✓	Engineering Unit Maximum
EU_Min	Input	REAL	✓	Engineering Unit Minimum
Analog_Ch	Input	REAL	✓	Analog Channel, Source
EU_Value	Output	REAL	✓	Scaled Engineering Unit Value

The **Local Tag** for the **SCL** instruction must be added with the **Data Type** of **SCALE**. Use the Tag name **Analog**.



Logic:

Type - **Structured Text**

SCL instruction is located in the **Process** tab.

See Handout, [Studio 5000 ANALOG SCALING - FUNCTION BLOCK and STRUCTURED TEXT](#).

Edit the example from the handout [Studio 5000 ANALOG SCALING - FUNCTION BLOCK and STRUCTURED TEXT](#) to use the **Tag** name **Analog** and **not Analog_In**. The Tags assigned with the **Analog.In** and **Analog.Out** are also different that the names used in the example.

continue

MainRoutine

The Routine **Scale** for scaling the analog input is called unconditionally.

The Routine **ConveyorControlRoutine** for the parts stack is called unconditionally.

Name of Routine: Scale

Tag for Add On Instruction HEIGHT_SCALE.

Use the Add On Instruction with these Constant Value and the described Tag name.

Parameter Name	Constant or Tag
Raw_Max	32767
Raw_Min	0
EU_Max	100
EU_Min	0
Analog_Ch	Tag for Height Sensor
EU_Value	Tag for Scaled EU Value

Name of Routine: ConveyorControlRoutine

See above explanation for logic requirements.

DOWNLOAD FOR INITIAL TESTING

Download the PLC program and place the PLC into RUN

*An HMI has been created to allow you to test the logic, **ANALOG_INPUT_HMI**.*

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 15 Appendix B: STATION RUNTIME APPLICATION.

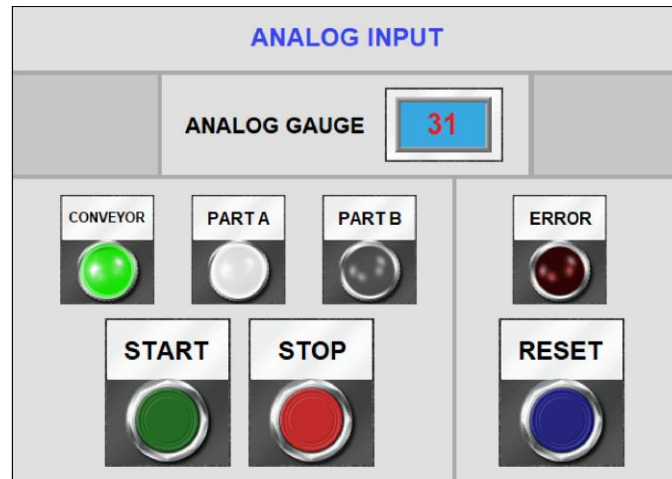
continued



TO TEST - HMI SCREEN

THIS HMI IS PROVIDED TO ALLOW YOU TO TEST THE PLC LOGIC.

YOU MUST STILL CREATE THE HMI LAB TO HAVE THE PLC LAB SIGNED OFF.



1. Press **START** to turn ON the CONVEYOR.

The Part Sequence is listed below.

PART A - Pass
PART B - Pass
PART A - Fail
PART B - Pass
PART A - Pass
PART B - Fail

Pattern of six parts repeats.

2. When an ERROR occurs, press **RESET**.
3. After resetting, press **START** to turn ON the CONVEYOR.

After test for the PART A ERROR and PART B ERROR check the STOP Button.

1. Press **START** to turn ON the CONVEYOR.
2. Press **STOP** to turn OFF the CONVEYOR.

continue

DOWNLOAD

Test the PLC program with the HMI to determine if there are problems with the HMI settings.

Test using the HMI you created for the HUMAN MACHINE INTERFACE (HMI) LABS the ANALOG INPUT lab.

Download the PLC program and place the PLC into RUN and execute the HMI in Station Runtime.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 15 Appendix B: STATION RUNTIME APPLICATION.

TESTING

START

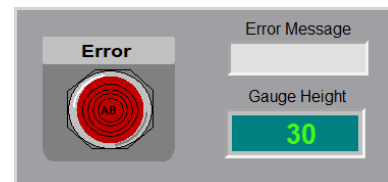
1. Press the **START** button.
CONVEYOR should be ON.



TESTING FOR A SINGLE PART A

In the PLC program look for the Input PART_A to be TRUE.

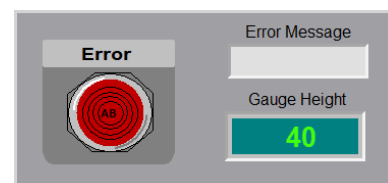
There should be No Error.



TESTING FOR A SINGLE PART B

In the PLC program look for the Input PART_B to be TRUE.

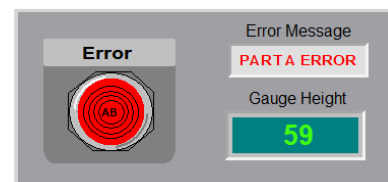
There should be No Error.



TESTING FOR A DOUBLE PART A

In the PLC program look for the Input PART_A to be TRUE.

The ERROR indicator should be ON.



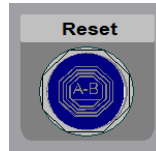
continue



RESET ERROR

1. Press the **RESET** button.

The **ERROR** indicator turns OFF.



START CONVEYOR

1. Press the **START** button.

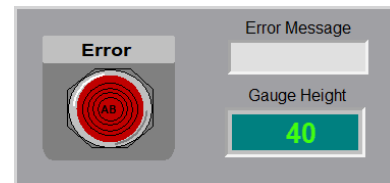
CONVEYOR should be ON.



TESTING FOR A SINGLE PART B

In the PLC program look for the Input PART_B to be TRUE.

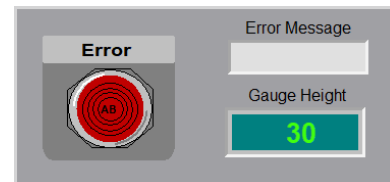
There should be No Error.



TESTING FOR A SINGLE PART A

In the PLC program look for the Input PART_A to be TRUE.

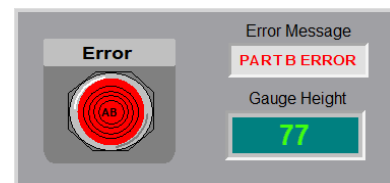
There should be No Error.



TESTING FOR A DOUBLE PART B

In the PLC program look for the Input PART_B to be TRUE.

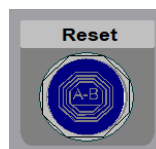
The **ERROR** indicator should be ON.



RESET ERROR

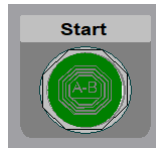
1. Press the **RESET** button.

The **ERROR** indicator turns OFF.



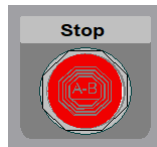
START CONVEYOR

1. Press the **START** button.
CONVEYOR should be ON.



STOP CONVEYOR

1. Press the **STOP** button.
CONVEYOR should be OFF.



LAB: TORQUE

PROGRAM FILE: TORQUE

OBJECTIVE: Use of multiple analog input channels.

NOTE: The scaling is performed by the Analog Input Module, the instructions for scaling are not required. Data from the module AIN_1, AIN_2, AIN_3, and AIN_4 can be used directly in the program's logic.

- The operator places a nut onto four bolts and moves the part into a fixture.
- The operator will then press and hold the two momentary normally open Start pushbuttons which energize the Down Solenoid. The pushbuttons are pressed for the entire operation.
- The Down Solenoid is energized, and the nut runner carriage is lowered. Turning off the down solenoid will raise the nut runner.
- When the nut runner carriage travel hits the Down Limit switch, the air motors are energized and will automatically tighten the nuts.
- A Torque Sensor is attached to each nut runner.
- When the torque reaches 95 to 100 foot pounds, the individual motor will turn off.
- When all four motors are de-energized, the Down Solenoid is de-energized and the nut runner carriage returns to the up position.

Torque Timeout Error

- A time out is included, so that after a total of 20 seconds of operations if torque is not reached by a nut runner motor. The following will occur:
 - The nut runner motors, and the Down Solenoid will turn Off.
 - An error output, TO_ERROR, will be turned on.
- When both START pushbuttons are released, it will reset the time out error.

The release of either START pushbutton will turn off the Down Solenoid and the nut runner motors. The last torque value is retained by the torque sensor.

The torque sensors are reset by the torque sensor transducer.

The upper and lower limits for the torque are defined in tags and are not hard coded constants.

I/O CONFIGURATION:

Slot 2 - Studio Emulate 5000 Controller

Slot 3 - 1756-IB16

Slot 4 - 1756-OB16E

Slot 5 - 1756-IF8

NOTE: Tags which begin with **em** *TagName* are for the emulation model and are not part of the program.

INPUTS:

Alias Address	Controller Tag	Descriptions
Local:3:I.Data.0	DN_LS	Down Travel N. O. Limit Switch
Local:3:I.Data.14	START_NO_1	START No. 1 Pushbutton
Local:3:I.Data.15	START_NO_2	START No. 2 Pushbutton

OUTPUTS:

Alias Address	Controller Tag	Descriptions
Local:4:O.Data.0	DN_SOL	Down Solenoid
Local:4:O.Data.1	MOTOR_1	Nut Runner No. 1, Air Motor
Local:4:O.Data.2	MOTOR_2	Nut Runner No. 2, Air Motor
Local:4:O.Data.3	MOTOR_3	Nut Runner No. 3, Air Motor
Local:4:O.Data.4	MOTOR_4	Nut Runner No. 4, Air Motor
Local:4:O.Data.10	TO_ERROR	Time Out Indicator

ANALOG INPUTS:

Analog Data are REAL Data Types by default.

Alias Address	Controller Tag	Descriptions
Local:5:I.Ch0Data	AIN_1	Channel 1, Torque Sensor No. 1
Local:5:I.Ch1Data	AIN_2	Channel 2, Torque Sensor No. 2
Local:5:I.Ch2Data	AIN_3	Channel 3, Torque Sensor No. 3
Local:5:I.Ch3Data	AIN_4	Channel 4, Torque Sensor No. 4



INTERNALS:

Controller Tag	Data Type	Descriptions
UPPER_LIMIT	DINT	Upper Torque Setting
LOWER_LIMIT	DINT	Lower Torque Setting

Param/Local Tag	Data Type	Descriptions
TORQUE_1_SET	Boolean	No. 1 Bolt Torque Set
TORQUE_2_SET	Boolean	No. 2 Bolt Torque Set
TORQUE_3_SET	Boolean	No. 3 Bolt Torque Set
TORQUE_4_SET	Boolean	No. 4 Bolt Torque Set

TIMERS:

(1) The tags for the status bits DN, TT, and EN are created automatically when timer tag is created.

Param/Local Tag	Data Type	Descriptions
TO_DWELL	Timer	Time Out Dwell
TO_DWELL.DN ⁽¹⁾	Boolean	Set Time Out Error (DN)
TO_DWELL.TT ⁽¹⁾	Boolean	Time Out Elapsing (TT) <i>optional</i>
TO_DWELL.EN ⁽¹⁾	Boolean	Time Out Enabled (EN) <i>optional</i>

DOWNLOAD

Download the PLC program and place the PLC into RUN

An HMI has been created to allow you to test the logic, TORQUE_HMI.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 15 Appendix B: STATION RUNTIME APPLICATION.

INFORMATION - HMI SCREENS

Two HMI screens are used in the running and testing of the lab.

The first screen is the MAIN screen with the two START pushbuttons. When a pushbutton is depressed, such as START 1, the button color is a darker green.

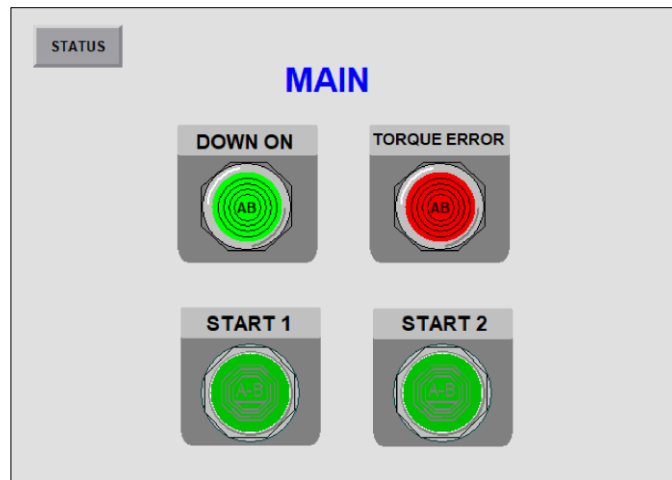
continue



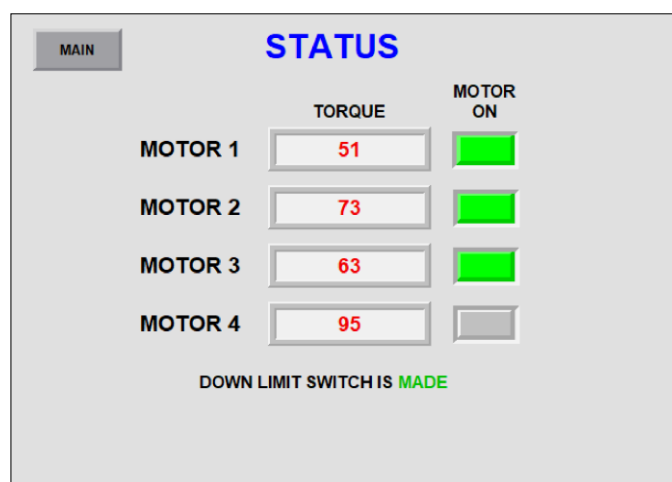
NOTE: START 1 and START 2 are Maintained Pushbuttons in the HMI to allow you to change to the Status Screen.

After a cycle, release both the START pushbuttons by putting the cursor on the pushbutton and left clicking.

An indicator is illuminated bright green for the DOWN ON solenoid. The second indicator illuminates bright red when a torque error is generated. A navigation button in the upper left goes to the second screen, STATUS.



The second HMI screen STATUS has four indicators for each motors illuminating bright green when the motor is ON. The torque value of each motor is displayed. At the bottom of the screen the text informs if the Down Limit Switch is MADE (closed) or APART (opened). A navigation button in the upper left returns to the first MAIN screen.

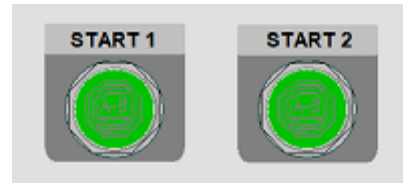


continue

PART 1 AND PART 2

In the PLC program, check that the upper and lower torque limits are set to a LOW LIMIT of 95 and a HIGH LIMIT of 100.

1. Place the cursor on the **START 1** pushbutton and left click
2. Place the cursor on the **START 2** pushbutton and left click.



- The output for the **DOWN SOLENOID** should turn ON.



3. Select the go to button **STATUS**.



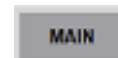
- Next the limit switch for the down travel will be set and the nut runner motors should turn ON.
- The Analog Input values from the Torque Sensors AIN_1, AIN_2, AIN_3, and AIN_4 will be set by the environment to simulate the torque signals values from the individual nut runner motor.
- When all four nut runners have reached torque the **DOWN SOLENOID** will turn OFF.
- The following pattern will be repeated each time a cycle is Started:

	TORQUE	MOTOR ON
MOTOR 1	51	
MOTOR 2	73	
MOTOR 3	63	
MOTOR 4	95	
DOWN LIMIT SWITCH IS MADE		

Part 1 - All motors reach torque.
Part 2 - All motors reach torque.
Part 3 - Motor 1 fails to reach torque.

Pattern of three parts repeats.

4. When all Motors are OFF, select the go to button **MAIN**.



5. Release both the **START** pushbuttons by putting the cursor on the pushbutton and left clicking.
 - The TIME OUT timer is reset.

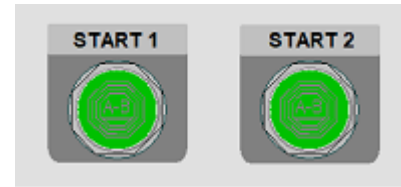
Repeat Step 1 through 5 for Part 2.

continue

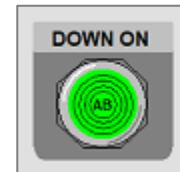


PART 3 - MOTOR 1 ERROR

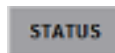
1. Place the cursor on the **START 1** pushbutton and left click
2. Place the cursor on the **START 2** pushbutton and left click.



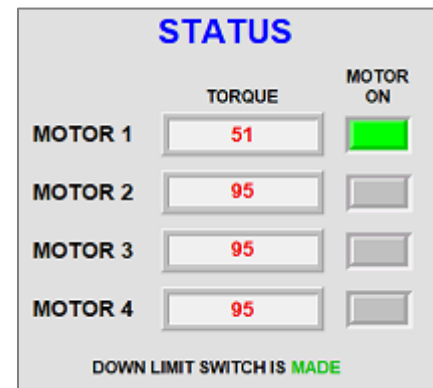
- The output for the **DOWN SOLENOID** should turn ON.



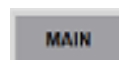
3. Select the go to button **STATUS**.



- Next the limit switch for the down travel will be set and the nut runner motors should turn ON.
- The Analog Input values from the Torque Sensors will turn OFF nut runner **MOTOR 2**, **MOTOR 3**, and **MOTOR 4**.
- In the PLC program notice the timer for the TIME OUT is elapsing.
- When the **Timer Times Out** the nut runner **MOTORS** still on will turn OFF.
- The **DOWN SOLENOID** turns OFF.



4. Select the go to button **MAIN**.



- The **TORQUE ERROR** indicator is ON.



5. Release both the **START** pushbuttons by putting the cursor on the pushbutton and left clicking.
 - The **TORQUE ERROR** indicator turns OFF.
 - The **TIME OUT** timer is reset.

